



JSS
SCIENCE AND
TECHNOLOGY
UNIVERSITY
MYSURU

- Constituent College of JSS Science and Technology University
- Approved by A.I.C.T.E
- Governed by the Grant-in-Aid Rules of Government of Karnataka
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DEPARTMENT OF MECHANICAL ENGINEERING

I / II Semester B.E. Degree

Elements of Mechanical Engineering [22ME110/22ME210]

Faculty Name: Karthik MS

QUESTION BANK

UNIT - 1

Steam Formation:

- (1) Explain the formation of steam at constant pressure with suitable sketches and T-H diagram.
- (2) Define: (a) Saturated steam (b) wet steam (c) dry steam (d) Super-heated steam
(e) Enthalpy of: - wet steam, dry steam and Super-heated steam
(f) Specific volume of: - wet steam, dry steam and Super-heated steam
(g) Internal energy (h) external work of evaporation. (i) Dryness fraction
- (3) Define: (i) sensible heat (ii) latent heat (iii) super heat (iv) degree of super heat.

Boilers:

- (1) Define boilers, classification of boilers.
- (2) Explain with a neat sketch water tube boiler. List out the industrial water tube boilers.
- (3) Explain with a neat sketch fire tube boiler. List out the industrial fire tube boilers.
- (4) List out the difference between fire tube and water tube boiler.
- (5) Explain briefly the following boiler mounting.
 - i. Safety valve
 - ii. Water level indicator
 - iii. Pressure gauge
 - iv. Steam stop valve
 - v. Blow off valve
- (6) Explain briefly the following boiler accessories.
 - (i) Economizer (ii) Air pre heater (iii) Super heater
 - (iv) Feed pump (v) Steam separator (vi) Steam trap